Instituto Politécnico Nacional
Centro de Investigación en Computación

Mathematics for Computer Science

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Homework 2

For the function $f(x) = \dfrac{7-2x}{5x-4}$, find domain of $f(x)$, $f^{-1}(x)$ and domain of $f^{-1}(x)$.

Solution

1. $f(x)$ is defined for every real number, except for $5x - 4 = 0$. Therefore the domain of $f(x)$ is the set of all real numbers such that $x \neq \dfrac{4}{5}$.

2. To find $f^{-1}(x)$, we substitute $y = f(x)$

$$\begin{aligned} y &= \frac{7-2x}{5x-4} \\ y\,(5x-4) &= 7-2x \\ 5xy-4y &= 7-2x \\ 5xy+2x &= 7+4y \\ x\,(5y+2) &= 4y+7 \\ x &= \frac{4y+7}{5y+2} \end{aligned}$$

   Therefore $f^{-1}(x) = \dfrac{4x+7}{5x+2}$.

3. The domain for this inverse function is the set of all the real numbers, except $x = -\dfrac{2}{5}$.